

TUNABLE NON-LINEAR MANIFOLD ROMs FOR ELIPTIC, PARABOLIC, AND HYPERBOLIC PDES VIA MATRIX-FREE PETROV–GALERKIN PROJECTION

Anonymous authors

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ABSTRACT

Learned reduced models are usually deployed with a fixed approximation space. Solving their equations more accurately cannot remove the error imposed by that space. We present a nonlinear-manifold reduced-order model (NM-ROM) for Poisson, heat and viscous Burgers equations that exposes a deployment-time accuracy–cost choice from a single trained decoder. The correction rank controls the size of the trial space, while solver tolerance and empirical quadrature control the cost of the reduced solve. The framework combines a learned spatial bank and nonlinear coefficient head with nested correction directions, exact Dirichlet enforcement, weak Petrov–Galerkin residual minimization, and a matrix-free JAX implementation with preassembled linear terms. On two-dimensional Burgers, increasing the correction rank lowers error at increased cost with the test space held fixed. On Poisson and heat, analytical elimination of the corrections improves accuracy at nearly constant cost. At 4096^2 , measured speedups over the named iterative solvers reach $116\times$ at 0.96 % error on Poisson and $36.0\times$ at 0.49 % on held-out heat. Held-out Burgers reaches 1.33 % at $5.38\times$, although its quadrature rule fails a subsequent confirmation check. The gains depend on the problem and comparator: linear solves in the same heat bank are faster, and the method misses its targets on three-dimensional Burgers, Navier–Stokes and full-state waves.

1 INTRODUCTION

Neural operators for partial differential equations (PDEs), including Fourier Neural Operators and DeepONets (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023), learn fast input-to-solution maps. Their accuracy and cost depend on training and deployment choices, including evaluation resolution. Classical full-order methods (FOMs) expose an accuracy–cost choice through the stopping tolerance of an iterative solver, but their cost grows with the mesh. We seek a reduced PDE solver whose approximation capacity can also be chosen at deployment, without retraining its decoder.

Reduced-order models (ROMs) approach this problem through a smaller solution space. Linear projection-based ROMs (Benner et al., 2015; Sirovich, 1987) can require many basis functions to represent moving fronts (Cohen and DeVore, 2015). Nonlinear-manifold ROMs learn a more compact representation (Lee and Carlberg, 2020; Kim et al., 2022), but the online nonlinear solve can offset the benefit of fewer unknowns. Moreover, tightening its tolerance cannot remove approximation error from a fixed manifold. The key question is thus: *can one trained NM-ROM provide several accuracy levels, retain its accuracy under mesh refinement, and offer useful runtime savings over iterative full-order solvers?*

We present an NM-ROM whose distinguishing property is a *deployment-time accuracy–cost choice from a single trained model*. A learned spatial bank supplies the basis functions; a small nonlinear head maps latent variables to their coefficients. Nested correction directions extend those coefficients after training. At inference, the user selects the number of active directions, the *correction rank* q , and the weak PDE residual determines their coefficients together with the latent state. All trained weights remain fixed.

Burgers exhibits an accuracy–cost tradeoff as correction rank grows; Poisson and heat allow analytical elimination of corrections, improving accuracy at nearly constant cost. Under two-dimensional mesh

refinement, errors remain similar while speedups over the named solvers grow. Linear-bank heat solves remain faster, and the high-resolution Burgers rule fails a later quadrature confirmation check.

Contributions.

1. **A tunable NM-ROM from a single trained decoder.** Nested corrections provide discrete deployment-time accuracy settings. We isolate their effect at fixed test count and test a separate rank-dependent schedule across seeds and held-out cases.
2. **Architectural choices that support this tradeoff.** A coordinate-network bank, nonlinear head and nested corrections allow compact latent solves, exact boundary enforcement and evaluation at selected quadrature nodes.
3. **A matrix-free JAX implementation.** Preassembled linear terms and empirical quadrature reduce online work. Comparisons with named iterative FOMs measure accuracy and runtime and identify where the method fails.

2 RELATED WORK

Linear and non-linear manifold ROMs. Linear ROMs approximate solutions in a fixed subspace (Benner et al., 2015; Sirovich, 1987); moving fronts can require a large basis (Cohen and DeVore, 2015; Greif and Urban, 2019). Nonlinear-manifold ROMs combine learned representations with residual projection (Lee and Carlberg, 2020; Kim et al., 2022), including quadratic manifolds (Geelen et al., 2022; Barnett and Farhat, 2022), neural-field representations (Kim et al., 2024) and adaptive bases (Peherstorfer and Willcox, 2015; Carlberg, 2015). Least-squares Petrov–Galerkin projection (Carlberg et al., 2011) and preassembly of reduced operators (Ştefănescu and Sandu, 2014; Weder et al., 2024) are established components. We differ from these by a coordinate-network bank with an exact boundary factor, a head with a linear skip, and nested corrections selected after training that give deployment-time settings from one trained model, as runtime-tunable networks do (Yu et al., 2019; Cai et al., 2020).

Neural operators and hyper-reduction. FNO (Li et al., 2021), DeepONet (Lu et al., 2021), U-Net (Ronneberger et al., 2015; Takamoto et al., 2022) and Transolver (Wu et al., 2024) learn solution maps. Their evaluation resolution or inference procedure can also change accuracy and cost; our mechanism instead changes the trial space used to solve the PDE. Empirical quadrature (Hernández et al., 2017; Yano and Patera, 2019) reduces nonlinear residual evaluation through non-negative weights. We retain that construction, testing its error on states reached by the solver and through independent re-draws. The main iterative linear-system comparator is conjugate gradient (Hestenes and Stiefel, 1952).

3 METHODOLOGY

Offline, we train the decoder, construct corrections and preassemble linear operators. Online, we choose a correction rank, solve for the reduced coordinates and reconstruct the requested fields (Figure A.1). The formulas below use a scalar Dirichlet problem on a grid with N intervals per axis, n interior unknowns and discrete negative Laplacian $A \in \mathbb{R}^{n \times n}$. Appendix A gives the per-PDE details.

3.1 TRIAL MANIFOLD WITH EXACT DIRICHLET ENFORCEMENT

The decoder represents a field as a weighted sum of R spatial functions. These functions form the columns of the frozen *bank*, $G \in \mathbb{R}^{n \times R}$. The *head*, $h_\theta : \mathbb{R}^k \rightarrow \mathbb{R}^R$, maps a small latent state z to their coefficients, with $k \leq R \ll n$. There is no encoder in the deployed solver. We build the bank once per mesh by evaluating a random-Fourier-feature coordinate network g_ϕ (Tancik et al., 2020). On the unit square, we multiply each spatial function by a factor that vanishes at the boundary:

$$G_{x,:} = \mu(x) g_\phi(x)^\top, \quad \mu(x) = 16 x_1(1 - x_1) x_2(1 - x_2). \quad (1)$$

Because every bank function is zero on the boundary, every decoded field satisfies homogeneous Dirichlet conditions, whatever its coefficients. The cube uses a product factor and the L-shape a distance-based factor; periodic Navier–Stokes uses no boundary factor (§3.4).

Nested correction directions. The head alone can produce only a restricted set of bank coefficients. We extend this set with q fixed directions, collected in $C_q \in \mathbb{R}^{R \times q}$. The online solver adjusts both the latent state z and the correction coefficients y :

$$u(z, y) = G(h_\theta(z) + C_q y), \quad z \in \mathbb{R}^k, y \in \mathbb{R}^q, 0 \leq q \leq R. \quad (2)$$

We choose the directions offline from training data. For Burgers, they are principal components of the head’s coefficient errors; other constructions are given in Appendix A. The directions are ordered: C_q contains the first q columns of any larger $C_{q'}$. Increasing q therefore enlarges the trial space without retraining.

At $q = 0$, only the head is active. At $q = R$, the corrections span the entire bank, so the head is redundant. Between these endpoints, the solver has $k + q$ coordinates. Linear PDEs allow the correction coefficients to be eliminated, as described below; Burgers requires a joint nonlinear solve.

All settings remain in the span of G . The bank’s projection error is therefore a lower bound on the error any setting can attain. The head can reduce the number of coordinates needed to represent a field, but it cannot represent fields outside that span.

3.2 PROJECTING THE PDE ONTO THE MANIFOLD

A discrete PDE defines a residual $r(u)$ that vanishes at its full-order solution. We evaluate this residual at the decoded field and measure its components along M fixed, smooth test functions. This is the *weak residual*: it tests the PDE against selected spatial modes rather than minimizing every grid-point residual separately.

The test matrix $P \in \mathbb{R}^{M \times n}$ contains low-frequency sine modes on the square. These are eigenvectors of the discrete Laplacian, so $PA = \Lambda P$, with Λ holding their eigenvalues. On the L-shape we compute eigenvectors of that domain’s operator. The online problem is

$$(z^*, y^*) = \arg \min_{z, y} \frac{1}{2} \|r_w(z, y)\|_2^2, \quad r_w(z, y) = \Lambda_\star^{-1} P r(u(z, y)) \in \mathbb{R}^M, \quad M > k + q, \quad (3)$$

where $\Lambda_\star$ scales the residual components as specified for each PDE in Appendix A. We use more tests than solved coordinates. This overdetermined least-squares problem defines our Petrov–Galerkin projection (Carlberg et al., 2011; Lee and Carlberg, 2020). The test functions stay fixed during the solve.

Poisson. For $Au = f$, the scaled weak residual is $r_w(z, y) = B_0(h_\theta(z) + C_q y) - b_0$, where $B_0 = PG$. For any fixed z , this is linear in y . We can therefore solve for y exactly and substitute it back into the objective (Golub and Pereyra, 1973). The remaining nonlinear iteration has only k variables, regardless of the correction rank. At $q = R$, the entire problem becomes a reduced linear solve (Appendix A.1).

Heat. We first discretize $du/dt = -\kappa Au$ with Crank–Nicolson, then substitute the decoder into that discrete equation and project. Each step fits the next decoded state to the propagated weak moments of the previous decoded state. The head makes this a nonlinear least-squares problem, but the correction coefficients can again be eliminated exactly. At $q = R$, the step is a linear recurrence in bank coefficients (Appendix A.2). Reflective waves use a separate explicit latent integrator (Appendix A.4).

Burgers. For $u_t + u(u_x + u_y) = \nu \Delta u$, we use backward Euler and the same sign-upwind advection stencil as the FOM. Advection is nonlinear in the bank coefficients, so we solve for z and y together. A Levenberg–Marquardt (LM) step damps the latent and correction blocks separately. Appendix A.3 gives the residual and update equations.

Initialization and stopping. Poisson starts from the stored training code nearest the projected source. Heat and Burgers start from stored codes and fit the decoded field to the supplied initial condition. These steps use the PDE input, without access to the solution being predicted.

The nonlinear solves use damped LM (Marquardt, 1963). A proposed step is accepted only if it decreases the residual norm. For Poisson and Burgers, stationarity is measured by

$$\eta(z, y) = \frac{\|J^\top r_w\|_2}{\|J\|_F \|r_w\|_2} \leq \eta_{\text{tol}}, \quad (4)$$

where J is the residual Jacobian. We distinguish this exit from reaching a residual threshold, stalling, or exhausting the iteration budget. A stall does not establish convergence. The heat normalization and all exit codes are given in Appendix A.6.

3.3 PREASSEMBLY AND EMPIRICAL QUADRATURE

A small number of unknowns does not by itself make a solve cheap: evaluating the PDE on the full grid can still dominate runtime. The bank lets us precompute every fixed linear term. For example, $PAG = \Lambda PG$ is an $M \times R$ matrix assembled once per mesh. Online, multiplying this matrix by the bank coefficients replaces applying the full-grid linear operator. This preassembly is exact.

Burgers advection still requires field values and their neighboring stencil values. Dense evaluation costs $O(nR)$ per residual. Empirical quadrature (EQ) approximates its weak projection using m selected nodes and non-negative weights (Hernández et al., 2017; Yano and Patera, 2019). Cached bank rows at those nodes and their neighbors reduce the spatial evaluation cost to $O(mR)$. We fit the weights offline by non-negative least squares (NNLS) (Lawson and Hanson, 1974) on decoded training states. Each stored rule has fixed nodes and weights; changing m requires a new fit, and the rules are not nested.

A small fitting error is insufficient to validate a rule. We measure its relative error ρ in projected advection on states reached by the solver, using trajectories separate from fitting and evaluation. The primary threshold is $\rho_{\max} \leq 0.116$, fixed before the validation runs, with a tighter threshold of 0.06. We then repeat the rule construction with independent draws. A *confirmed* rule passes every required draw, including the confirmation check. This is an empirical test over sampled states, not a global error bound (Appendix A.5).

3.4 MODEL ARCHITECTURE

Head. $h_\theta(z) = \varphi_\theta(z) + W^\top z$, where φ_θ is an MLP with two hidden SiLU layers. The linear skip supplies a latent-independent Jacobian component; its effect is not ablated here. Widths and reduced dimensions are listed in Tables B.2 and B.6.

Spatial bank. The coordinate network’s parameters are independent of the mesh. Quadrature uses cached rows, and decoding in blocks limits memory at large grids. Some banks use a fixed orthonormalising matrix. The periodic Navier–Stokes bank uses a global solenoidal projection. Boundary factors are described in §3.1.

Deployment settings. We choose q , the test count M , a stored quadrature rule, and solver tolerance and budget. Usually $M = 4(k + q)$; it stays fixed in the rank-isolation study and on the L-shape. Heat also supports the batched fit described in §6.1. None of these choices changes the trained weights.

4 IMPLEMENTATION

4.1 MATRIX-FREE PROJECTED OPERATORS VIA JAX

For a linear residual $B(h_\theta(z) + C_q y) - b$, the Jacobian is $B[Dh_\theta \ C_q]$. JAX forward-mode differentiation computes the head derivatives (Bradbury et al., 2018); the Burgers residual also uses cached bank rows at the sampled stencils. We form and solve the small damped normal system directly. “Matrix-free” refers to avoiding full-order Jacobian assembly; it does not mean that the reduced normal matrix is implicit. Burgers runs above 1024^2 use an analytical Jacobian and the solver refinements in Appendix A.3.

4.2 TRAINING PROTOCOL

We train the bank first and then fit the head: the coordinate network g_ϕ is fitted to the training states through (1) and frozen. The head and a library of training codes Z are then fitted to the bank-projected states; C_q is then constructed from training-only coefficient directions as specified for each PDE. Sizes and cohorts are in Appendix B; checkpoint hashes and recorded offline costs are retained with the source evidence described there.

5 EXPERIMENTAL SETUP AND BENCHMARK PROBLEMS

Problems and references. The two-dimensional problems are Poisson on the square and on an L-shaped domain, heat, viscous Burgers and reflective waves; the three-dimensional problems are Poisson, heat, Burgers and incompressible Navier–Stokes. Table B.2 gives the two-dimensional meshes and cohorts, Table B.3 the sampled families and Table B.6 the three-dimensional configurations. Burgers provides the correction-rank study.

Accuracy. We report worst relative L^2 error over the stated cases and requested times, in percent. Poisson uses the steady reference norm and heat the current reference norm; Burgers and Navier–Stokes use the initial-field norm over evolved times; waves use the energy-state error over displacement and velocity. Same-grid error measures reduction error relative to the converged discrete solution. Error against a refined reference also includes discretisation error; the two are never interchanged. Settings for a held-out (final) cohort are fixed before its cases are accessed; all other cohorts are development cohorts.

Timing and speedup. Accuracy and runtime are measured in the same solver invocation. We use double precision, GPU burn-in, synchronised timing and medians of retained repetitions. GPU query time includes initialization, the solve or rollout, and requested device outputs. Complete-query time additionally includes host transfers and is labelled separately. Training and compilation are offline costs. Speedup is $S = T_{\text{FOM}}/T_{\text{method}}$ with both times from the same GPU allocation. Each table names the FOM algorithm and shows its error. Solves that miss their stopping rule are reported as such and do not enter a speedup.

Baselines. Linear problems are compared with conjugate gradients (CG), including the L-shaped domain and the CG solve inside each Crank–Nicolson (CN) heat step. Burgers is compared with Newton–BiCGStab and Navier–Stokes with its CNAB2 time integrator. The named FOM of a row is that solver at its fastest tested setting that is at least as accurate as the accurate NM-ROM setting.

6 NUMERICAL EXPERIMENTS AND RESULTS

We evaluate three questions: whether corrections improve accuracy without retraining, when the reduced solve is faster than the named full-order solver, and whether the results carry to unseen cases. Table 1 gives the overall comparison; the rank studies separate the effect of corrections from changes in the test space.

6.1 ACCURACY AND SPEED AGAINST FULL-ORDER SOLVERS

The clearest test of the accuracy control holds the model and test space fixed. On Burgers at 256^2 , increasing q from zero to the top rank reduces error by $2.44\times$ and increases runtime by $5.16\times$ (Appendix Table B.5). Every setting meets its stopping rule. Thus the improvement comes from the added correction directions rather than a larger test space or a shorter, unfinished solve. The remaining experiments ask what this control offers at deployment.

Poisson and heat. On Poisson, increasing the correction rank improves accuracy with little change in runtime because the correction coefficients are eliminated analytically. The accurate setting stays near 0.96 % error as the mesh grows to 4096^2 , where its GPU-query speedup reaches $116\times$ over CG. The L-shaped problem also gives a speedup ($6.07\times$ at 2.12 % error).

Table 1: NM-ROM against the named full-order solver at every measured problem and mesh. **Both NM-ROM columns of a row come from one frozen model:** *fast* is the uncorrected setting ($q = 0$) and *accurate* the selected accurate rank (Table B.4); no network is retrained between them. Error is the worst relative L^2 error over the cohort (%). Speedup is the FOM’s median time divided by the NM-ROM’s in the same GPU allocation, **bold** where the NM-ROM is faster. The FOM is the named iterative solver at its fastest tested setting that is at least as accurate as the accurate setting. ^f held-out final cohort (all others development); ^r error against a refined reference, which includes discretisation error (all others same-grid); ^t all output times including $t = 0$; ^e evolved output times only; ^c complete-query time with host transfers (all others GPU query; heat was measured in GPU query only). Burgers rule status: ^s confirmation failed; ^x refitted rule tested on one draw; ^ℓ lattice rule (small passing margin at 2048^2 ; confirmation failed at 4096^2); ^v passed the independent re-draw procedure; ^d dense residual. ⁿ a solve missed stationarity, so no speedup is reported. ^h held-out cases, separate from the sealed cohort. The earlier Burgers model may exit on a stall. Poisson (dev. sources) rows use development sources of a separate run and are not the held-out cohort. Dashes mark settings not measured. Times, settings, rule identities and complete-query speedups: Appendix B.1.

Problem	Mesh	NM-ROM accurate		NM-ROM fast		Full-order model (FOM)	
		Err. (%)	Speedup	Err. (%)	Speedup	Err. (%)	Solver
<i>Two-dimensional</i>							
Poisson	256 ²	0.97	3.13 ×	3.16	3.33 ×	0.15	CG, rtol 10 ^{−2}
	1024 ²	0.96	13.9 ×	3.15	15.2 ×	0.072	CG, rtol 10 ^{−2}
	2048 ²	0.96	72.8 ×	3.15	74.3 ×	0.049	CG, rtol 10 ^{−2}
	4096 ²	0.96	116 ×	3.15	117 ×	0.45	CG, rtol 10 ^{−1}
Poisson, L-shape ^c	256 ²	2.13	3.84 ×	3.86	4.04 ×	0.64	CG, rtol 10 ^{−2}
	512 ²	2.12	6.07 ×	3.85	6.18 ×	0.38	CG, rtol 10 ^{−2}
	1024 ²	2.20	8.38 ×	3.85	8.59 ×	1.05	CG, rtol 3×10 ^{−2}
	2048 ²	2.20	17.0 ×	3.85	17.1 ×	0.76	CG, rtol 3×10 ^{−2}
Heat ^r	64 ²	—	—	4.56	0.22×	1.59	CN-CG, rtol 10 ^{−2}
	256 ²	—	—	4.56	0.61×	0.93	CN-CG, rtol 10 ^{−2}
	1024 ²	—	—	4.56	4.85 ×	0.77	CN-CG, rtol 10 ^{−2}
Heat (wide bank) ^{t,f}	1024 ²	0.49	1.59 ×	1.36	1.57 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	2048 ²	0.49	9.74 ×	1.36	9.59 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	4096 ²	0.49	36.0 ×	1.36	35.4 ×	0.47	CN-CG, Δt=0.05, rtol 10 ^{−2}
Heat (wide bank, batched fit) ^{t,f}	1024 ²	0.49	13.9 ×	1.36	13.0 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	2048 ²	0.49	60.0 ×	1.36	57.0 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	4096 ²	0.49	103 ×	1.36	101 ×	0.47	CN-CG, Δt=0.05, rtol 10 ^{−2}
Burgers ^e	256 ²	0.51 ^s	0.043×	1.89	0.79×	0.049	Newton-BiCGStab, tol 10 ^{−3}
	512 ²	0.55 ^x	0.068×	2.14	1.31 ×	0.052	Newton-BiCGStab, tol 10 ^{−3}
	1024 ²	0.59 ^d	0.0037×	2.29	2.02 ×	0.034	Newton-BiCGStab, tol 10 ^{−4}
Burgers ^e	2048 ²	0.87 ^ℓ	0.99×	2.37	4.16 ×	0.049	Newton-BiCGStab, tol 3×10 ^{−3}
	4096 ²	0.60 ^ℓ	4.87 ×	2.41	12.9 ×	0.050	Newton-BiCGStab, tol 3×10 ^{−3}
Burgers (held-out cases) ^{e,h}	2048 ²	1.31 ^ℓ	1.43 ×	9.03	5.63 ×	0.13	Newton-BiCGStab, tol 10 ^{−3}
	4096 ²	1.33 ^ℓ	5.38 ×	9.03	13.2 ×	0.14	Newton-BiCGStab, tol 3×10 ^{−3}
Burgers, confirmed rule ^e	512 ²	0.56 ^v	0.091×	2.14	0.75×	0.050	Newton-BiCGStab, tol 3×10 ^{−3}
	1024 ²	0.59 ^v	0.32×	2.29	1.39 ×	0.048	Newton-BiCGStab, tol 3×10 ^{−3}
Burgers (earlier model) ^r	1024 ²	—	—	3.91	1.63 ×	2.39	Newton-BiCGStab, relaxed
<i>Three-dimensional</i>							
Poisson ^f	32 ³	0.26	0.94×	1.40	0.99×	0.16	CG, rtol 10 ^{−2}
	64 ³	0.26	1.33 ×	1.39	1.38 ×	0.11	CG, rtol 10 ^{−2}
Poisson (dev. sources)	128 ³	0.16	6.75 ×	0.55	6.98 ×	0.075	CG, rtol 10 ^{−2}
	256 ³	0.16	23.2 ×	0.55	23.7 ×	0.049	CG, rtol 10 ^{−2}
Heat (new bank) ^{t,f}	32 ³	0.11	0.057×	2.00	0.13×	0.078	CN-CG, Δt=0.025, rtol 10 ^{−4}
	64 ³	0.11	— ⁿ	2.00	0.21×	0.081	CN-CG, Δt=0.025, rtol 10 ^{−4}
	128 ³	0.11	0.28×	2.00	0.62×	0.082	CN-CG, Δt=0.025, rtol 10 ^{−4}
Heat (new bank, batched fit) ^{t,f}	32 ³	0.11	0.62×	2.00	0.57×	0.078	CN-CG, Δt=0.025, rtol 10 ^{−4}
	64 ³	0.11	0.94×	2.00	0.85×	0.081	CN-CG, Δt=0.025, rtol 10 ^{−4}
	128 ³	0.11	2.22 ×	2.00	2.10 ×	0.082	CN-CG, Δt=0.025, rtol 10 ^{−4}

On a sealed held-out heat cohort, the wide-bank model reaches 0.49 % error over all output times and is $36.0 \times$ faster than CN-CG at 4096^2 . Fitting each output independently to exactly propagated weak moments raises this to $103 \times$. This batched fit uses the linear, autonomous structure of heat and is not a general time integrator. A linear solve in the same bank is faster than either nonlinear setting (Limitations).

Table 2: Other nonlinear-manifold ROMs on the shared Burgers 2D family at 256^2 and 512^2 : worst and median evolved same-grid error over 32 validation cases (never used for any choice) and compiled-query memory, one allocation. Latent dimension is matched for the baselines and our fast setting; our accurate setting solves $k + q = 272$ unknowns. The shallow masked-autoencoder NM-LSPG of Kim et al. (2022) passed its reproduction gate on the third of three pre-registered attempts; its hyper-reduction was not reproduced, and its published encoder width does not fit our devices, so at 512^2 its width-capped encoder is training-time limited (Appendix B.2). Data-matched: trained on our bank’s 576 trajectories. The convolutional-autoencoder NM-ROM (Lee and Carlberg, 2020) was not run. No times: every row, ours included, uses a dense residual (Appendix B.2).

Method	Unknowns	256^2			512^2		
		Worst (%)	Median (%)	MB	Worst (%)	Median (%)	MB
Kim et al. NM-LSPG	8	163.93	43.34	2480	175.71	55.48	9959
Kim et al. NM-LSPG	16	145.40	43.87	2567	174.99	58.56	10511
Kim et al. NM-LSPG	32	127.51	43.61	2742	252.33	56.72	10879
Kim et al. NM-LSPG, data-matched	16	120.33	38.71	2567	—	—	—
POD-LSPG	8	55.52	22.72	18	55.82	24.08	71
POD-LSPG	16	41.53	10.45	30	42.01	13.83	122
POD-LSPG	32	24.78	5.65	55	25.63	7.74	222
This work, fast ($q = 0, k = 16$)	16	6.79	0.66	625	7.72	0.69	2368
This work, accurate ($q = 256, k = 16$)	272	0.88	0.089	1769	1.05	0.080	6891

In three dimensions, the held-out Poisson result is 0.26 % at $1.33\times$. The new heat bank reaches 0.11 % all-times error on 64 sealed cases. Its Crank–Nicolson rollout is slower than CN–CG at every mesh; only the batched fit is faster, at 128^3 ($2.22\times$).

Burgers. The uncorrected setting becomes faster than Newton–BiCGStab by 1024^2 , but its error is higher. Adding corrections improves accuracy and increases the solve cost. The accurate setting is slower than the FOM through 2048^2 , including the rules that pass independent re-draws at 512^2 and 1024^2 (rows marked ν). The dense 1024^2 panel also has a stored $q = 128$ quadrature rule that gives 1.04 % at $0.50\times$.

At 4096^2 , the accurate setting selected on six development cases reaches 0.60 % at $4.87\times$ the speed of the selected Newton solver. On held-out cases, error rises to 1.33 % at $5.38\times$. These are measured speedups, but the rule fails a subsequent quadrature confirmation check at this mesh. A passing variant with an exact first step loses the speed advantage. The separate bank follow-up also misses its combined held-out accuracy and speed target (Appendix B.1).

Effect of mesh resolution. The reduced models retain similar errors as the two-dimensional mesh is refined. Their query times grow more slowly than the iterative FOM times, so the speedups increase. The comparator setting is selected separately at each mesh by the stated accuracy rule; it need not be the same tolerance throughout a series. The corresponding query times are in Table B.4.

Comparison with other reduced models. On 32 validation Burgers cases at 256^2 and 512^2 (Table 2) our fast setting, at matched latent dimension, reaches worst errors of 6.79–7.72 %, against 120–252 % for the reproduced shallow masked-autoencoder NM-ROM and 25–56 % for POD-LSPG; the accurate setting, at 272 solved unknowns, reaches 0.88–1.05 %. This comparison concerns the reproduced configuration: encoder size and training time constrain the baseline, and its hyper-reduction was not reproduced.

Where the method currently fails. Table 3 reports the unsuccessful problems. Corrections improve three-dimensional Burgers, but the FOM remains faster and the refined-reference check fails; only same-grid reduction error is established. For Navier–Stokes, 31 of 32 held-out cases miss the 5 % target. A separate shift-tracking linear POD model handles the translating vortices more accurately (4.80 % worst error on a fresh cohort), but still runs at $0.47\times$ CNAB2. It is a different model. For waves, correcting the displacement and velocity improves energy-state error but removes the speed advantage. All three-dimensional residuals are evaluated densely.

Table 3: Problems on which the method misses its target: Navier–Stokes 5% per held-out case; wave, an energy-state error within that of CG; Burgers 3D, faster than the FOM at its accuracy (no numeric error target was recorded). Error is the worst same-grid relative L^2 error (%) over evolved times; the wave error is the energy-state error over displacement and velocity. Speedup is against the named FOM in the same allocation.

Problem	Setting	Err. (%)	FOM err. (%)	Speedup	FOM
Burgers 3D, 32^3 (final)	$q = 0$	16.26	2.31	$0.038\times$	Newton–BiCGStab
	$q = 192$	4.40	2.31	$0.023\times$	Newton–BiCGStab
Navier–Stokes 3D, 32^3 (final)	$q = 0$	20.34	2.47	$0.0025\times$	CNAB2
	$q = 256$	18.92	2.47	$0.0007\times$	CNAB2
Wave 2D, 1024^2 (dev.)	$q = 0$	11.34	4.29	$3.03\times$	midpoint–CG
	$q = 32$	5.12	4.29	$0.23\times$	midpoint–CG

Table 4: Correction-rank tunability at 4096^2 . Each problem block is one frozen model; every speedup is the median GPU query time of *one* named full-order setting (the *FOM* row, the fastest at least as accurate as the block’s most accurate setting, in the same allocation) divided by the setting’s (ms). Error is the worst same-grid relative L^2 error (%), over evolved times for Burgers and heat. Burgers development (6 cases) and held-out (64 cases, never used for selection) share settings; heat uses the sealed held-out cohort, Poisson development sources. On Burgers q and M change together ($M \approx 4(k + q)$, $M = 544$ an alternative at $q = 256$; fixed- M ladder: Table B.5), $q > 0$ with the lattice rule (ℓ in Table 1); heat and Poisson vary q only. * looser stopping tolerance; † quadrature rule failed its held-out check.

Setting	Development cohort			Held-out cohort		
	Err. (%)	ms	Speedup	Err. (%)	ms	Speedup
<i>Burgers 2D (6 development / 64 held-out cases)</i>						
$q = 0$, $M = 64$	2.41	40.7	$12.9\times$	9.03	40.6	$13.2\times$
$q = 128$, $M = 576$	1.09	77.4	$6.77\times$	2.43	74.6	$7.21\times$
$q = 256$, $M = 544$	0.88	112	$4.67\times$	2.93	105	$5.11\times$
$q = 256$, $M = 1088$	0.60	127	$4.12\times$	1.33	113	$4.77\times$
$q = 256$, $M = 1088$, looser tol.*	0.60	108	$4.87\times$	1.33	100	$5.38\times$
$q = 256$, $M = 2176^\dagger$	0.46	132	$3.96\times$	1.13	126	$4.25\times$
<i>FOM</i> : Newton–BiCGStab, tol. 3×10^{-3}	0.050	524	$1\times$	0.14	537	$1\times$
<i>Heat 2D, wide bank (16 sealed held-out cases)</i>						
$q = 0$	—	—	—	0.84	32.2	$146\times$
$q = 32$	—	—	—	0.22	31.7	$149\times$
$q = 96$	—	—	—	0.082	24.4	$193\times$
<i>FOM</i> : CN–CG, $\Delta t = 0.025$, rtol 10^{-6}	—	—	—	0.045	4713	$1\times$
<i>Poisson 2D (12 development sources)</i>						
$q = 0$	3.15	16.7	$117\times$	—	—	—
$q = 64$	2.08	16.9	$116\times$	—	—	—
$q = 128$	1.55	16.9	$116\times$	—	—	—
$q = 256$	0.96	16.9	$116\times$	—	—	—
<i>FOM</i> : CG, rtol 10^{-1}	0.45	1955	$1\times$	—	—	—

6.2 ACCURACY AND COST AS THE CORRECTION RANK CHANGES

Table 4 shows several settings of each frozen model. For Burgers, added corrections generally reduce error and require more work. Here the number of tests M changes with q , so this table measures the deployed settings together; the fixed- M experiment above isolates the effect of rank. Test count matters: with a smaller fixed test space, the same range of ranks reduces error by only $1.22\times$.

For Poisson and heat, the solver eliminates the linear correction coefficients analytically. Increasing q therefore improves accuracy at nearly constant cost. The labels “fast” and “accurate” identify the uncorrected and selected corrected settings; the uncorrected setting need not be faster on these linear problems.

Generalization across training seeds and cases. A separate dense Burgers study uses $M = 4(k + q)$ and evaluates the frozen settings once on a sealed cohort. Error decreases monotonically for

all 4 checkpoints; the top-rank errors are 0.59–0.68 % (Table C.1). Of these, 3 meet the pre-registered secondary criterion: at least a $2\times$ range in both error and cost. The stricter checks do not all pass. One sealed case reaches a wrong solution branch at $q = 0$, giving 10.1120 % error for the original model, and convergence fails for the second retrained seed at $q = 64$ (3 budget exits). Because M changes with q , this study does not replicate the fixed-test-space result.

Other deployment controls. Empirical quadrature and a looser stopping tolerance can reduce runtime with little change in error, within their tested ranges (Appendix Tables C.3 and C.2). They do not enlarge the representation. The iteration cap guards the runtime; stopping too early can leave a failed solve.

Limitations. (i) Speedups concern the named iterative FOMs; other solver classes (direct and spectral solvers, coarser discretisations) are outside this comparison. The FOM is more accurate in every row. Linear solves in the same heat bank are faster than every NM-ROM setting: 0.13 % in 5.1–8.0 ms at 4096², and 0.070 % in 1.5–2.8 ms at 128³. (ii) Bank and head capacity limit accuracy. Poisson errors are near the bank floors (0.742 % square, 0.143 % cube); the L-shape’s 2.20 % error exceeds its 0.77 % floor. Burgers accuracy worsens on held-out cases. (iii) Rank-only evidence uses one checkpoint. The multi-seed study changes the test count with rank, and FOM comparisons do not isolate the nonlinear head’s benefit over its bank. (iv) Quadrature checks cover finitely many states; none is used in three dimensions. A solve can reach a wrong branch, and low same-grid error does not imply low discretisation error. (v) Unless marked held-out or final, cohorts are small development sets. Timings report medians without dispersion.

7 CONCLUSION AND FUTURE WORK

Nested corrections let a trained decoder support several accuracy levels. They lower Burgers error at increased solve cost and improve Poisson and heat accuracy at nearly constant cost. The practical gain depends on representation quality, quadrature and the comparator: linear-bank heat solves remain stronger, the fast Burgers rule fails confirmation, and several three-dimensional and wave targets remain unmet. Further work should address these limits and reduce the cost of large correction spaces.

REPRODUCIBILITY STATEMENT

Every table and prose number is generated from hash-pinned run records that identify source, checkpoint, settings, cohort, allocation and audit (Appendix B); failed settings are retained.

AI USE STATEMENT

Language-model assistants supported method and experiment development, implementation, result analysis, auditing, and manuscript drafting and editing. Numerical results were computed by the recorded solver runs; table generators reproduce the reported values from those records. The authors are responsible for verifying the methods, results and text.

REFERENCES

- J. Barnett and C. Farhat. Quadratic approximation manifold for mitigating the Kolmogorov barrier in nonlinear projection-based model order reduction. *Journal of Computational Physics*, 464:111348, 2022.
- P. Benner, S. Gugercin, and K. Willcox. A survey of projection-based model reduction methods for parametric dynamical systems. *SIAM Review*, 57(4):483–531, 2015.
- J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. JAX: composable transformations of Python+NumPy programs, 2018.

- H. Cai, C. Gan, T. Wang, Z. Zhang, and S. Han. Once-for-All: Train one network and specialize it for efficient deployment. In *International Conference on Learning Representations*, 2020.
- K. Carlberg. Adaptive h -refinement for reduced-order models. *International Journal for Numerical Methods in Engineering*, 102(5):1192–1210, 2015.
- K. Carlberg, C. Bou-Mosleh, and C. Farhat. Efficient non-linear model reduction via a least-squares Petrov–Galerkin projection and compressive tensor approximations. *International Journal for Numerical Methods in Engineering*, 86(2):155–181, 2011.
- A. Cohen and R. DeVore. Approximation of high-dimensional parametric PDEs. *Acta Numerica*, 24:1–159, 2015.
- R. Geelen, S. Wright, and K. Willcox. Operator inference for non-intrusive model reduction with quadratic manifolds. *Computer Methods in Applied Mechanics and Engineering*, 403:115717, 2022.
- G. H. Golub and V. Pereyra. The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate. *SIAM Journal on Numerical Analysis*, 10(2):413–432, 1973.
- C. Greif and K. Urban. Decay of the Kolmogorov N -width for wave problems. *Applied Mathematics Letters*, 96:216–222, 2019.
- J. A. Hernández, M. A. Caicedo, and A. Ferrer. Dimensional hyper-reduction of nonlinear finite element models via empirical cubature. *Computer Methods in Applied Mechanics and Engineering*, 313:687–722, 2017.
- M. R. Hestenes and E. Stiefel. Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards*, 49(6):409–436, 1952.
- Y. Kim, Y. Choi, D. Widemann, and T. Zohdi. A fast and accurate physics-informed neural network reduced order model with shallow masked autoencoder. *Journal of Computational Physics*, 451:110841, 2022.
- M. Kim, T. Wen, K. Lee, and Y. Choi. Physics-informed reduced order model with conditional neural fields. *arXiv preprint arXiv:2412.05233*, 2024.
- N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- C. L. Lawson and R. J. Hanson. *Solving Least Squares Problems*. Prentice-Hall, Englewood Cliffs, NJ, 1974.
- K. Lee and K. T. Carlberg. Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders. *Journal of Computational Physics*, 404:108973, 2020.
- Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.
- L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.
- D. W. Marquardt. An algorithm for least-squares estimation of nonlinear parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963.
- B. Peherstorfer and K. Willcox. Online adaptive model reduction for nonlinear systems via low-rank updates. *SIAM Journal on Scientific Computing*, 37(4):A2123–A2150, 2015.
- O. Ronneberger, P. Fischer, and T. Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241, 2015.

- L. Sirovich. Turbulence and the dynamics of coherent structures, Parts I–III. *Quarterly of Applied Mathematics*, 45(3):561–590, 1987.
- R. Ștefănescu and A. Sandu. Efficient approximation of sparse Jacobians for time-implicit reduced order models. *arXiv preprint arXiv:1402.2018*, 2014.
- M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert. PDEBench: An extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 35, 2022.
- M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. T. Barron, and R. Ng. Fourier features let networks learn high frequency functions in low dimensional domains. *Advances in Neural Information Processing Systems*, 33, 2020.
- J. Weder, P. Schwerdtner, and B. Peherstorfer. Online-efficient model reduction on quadratic manifolds with Neural Galerkin schemes. *arXiv preprint arXiv:2412.17695*, 2024.
- H. Wu, H. Luo, H. Wang, J. Wang, and M. Long. Transolver: A fast transformer solver for PDEs on general geometries. In *International Conference on Machine Learning*, 2024.
- M. Yano and A. T. Patera. An LP empirical quadrature procedure for reduced basis treatment of parametrized nonlinear PDEs. *Computer Methods in Applied Mechanics and Engineering*, 344:1104–1123, 2019.
- J. Yu, L. Yang, N. Xu, J. Yang, and T. Huang. Slimmable neural networks. In *International Conference on Learning Representations*, 2019.

A METHOD DETAILS

The generic overdetermined problem (3) is instantiated differently per PDE. Poisson, heat and Burgers below are fully discrete weak least-squares problems; the wave arm (§A.4) is an explicit second-order latent integration and is not an instance of (3). Formulas are for the scalar two-dimensional Dirichlet case; three-dimensional and periodic-vector scope is stated in Appendix B.1.

A.1 ELLIPTIC INSTANCE: POISSON

For $Au = f$ we take $\Lambda_\star = \Lambda$, so the weak residual is

$$r_w(z, y) = \Lambda^{-1}P(Au - f) = B_0(h_\theta(z) + C_q y) - b_0, \quad B_0 = PG, \quad b_0 = \Lambda^{-1}Pf. \quad (5)$$

Because the retained tests are orthonormal and $Au^\star = f$, this residual is exactly the tested error $P(u - u^\star)$, so the minimised objective is the squared discrete L^2 error of the reduced state in the retained modes; this holds only for the modes P retains.

Analytically eliminated corrections. Let $B_0 C_q = Q\mathcal{R}$ be a thin QR factorisation. Q does not depend on z , so the inner minimisation over y has the closed form $\mathcal{R}y = Q^\top(b_0 - B_0 h_\theta(z))$ and the minimised value is

$$\min_y \|B_0(h_\theta(z) + C_q y) - b_0\| = \|B_\perp h_\theta(z) - b_\perp\|, \quad B_\perp = (I - QQ^\top)B_0, \quad b_\perp = (I - QQ^\top)b_0. \quad (6)$$

The LM iteration runs on z alone with the deflated operator, and y is recovered by one triangular solve. Because the projector is constant the elimination introduces no approximation and $B_\perp D h_\theta$ is the exact Jacobian of the eliminated residual; the only dropped derivative anywhere in the elliptic solve is the head curvature. At $q = R$ the deflated operator is zero, the latent problem is degenerate by construction, and the model is the linear least-squares solve $\min_y \|B_0 y - b_0\|$; the timed top rung is that QR solve, not an LM iteration on a zero Jacobian. We report the reduced stationarity of the deflated problem, the augmented stationarity of the full problem, the backward error of the triangular recovery and the numerical rank of $B_0 C_q$.

A.2 PARABOLIC INSTANCE: HEAT

The semi-discrete heat equation $\dot{u} = -\kappa Au$ is advanced with Crank–Nicolson, $(I + \frac{\Delta t \kappa}{2} A)u^{n+1} = (I - \frac{\Delta t \kappa}{2} A)u^n$, and the reduced step substitutes the manifold into the fully discrete equation *before* projecting. With $u^n = u(z_n)$, projecting with P and row-scaling by $\Lambda_\star = I + \frac{\Delta t \kappa}{2} \Lambda$ gives

$$r_{w,n}(z) = B_0 h_\theta(z) - D B_0 h_\theta(z_n), \quad D = \text{diag}\left(\frac{1 - \frac{\Delta t \kappa}{2} \lambda_i}{1 + \frac{\Delta t \kappa}{2} \lambda_i}\right), \quad (7)$$

and $z_{n+1} = \arg \min_z \|r_{w,n}(z)\|_2$, warm-started at z_n . The target is computed from the decoded current state at each step. With corrections, $c_n = h_\theta(z_n) + C_q y_n$ replaces $h_\theta(z_n)$:

$$(z_{n+1}, y_{n+1}) = \arg \min_{z,y} \|B_0(h_\theta(z) + C_q y) - D B_0 c_n\|_2, \quad (8)$$

where y is eliminated as in (6) and the *complete* previous coefficients c_n , corrections included, are carried into the next step. At $q = R$ the head is redundant and the step is the linear least-squares recurrence

$$c_{n+1} = B_0^+ D B_0 c_n, \quad (9)$$

with B_0^+ the pseudo-inverse ($M \geq R$; the numerical rank of B_0 is checked) and c_0 fitted to the supplied initial field. This endpoint is the weak Crank–Nicolson propagation of the bank coefficients, with no head and no iteration. The damped normal matrix is factored by Cholesky.

A.3 NONLINEAR ADVECTION: BURGERS

The full-order problem is $u_t + u(u_x + u_y) = \nu \Delta u$ with the non-conservative sign-upwind stencil

$$N(u)_{ij} = u_{ij} (\delta_x u + \delta_y u)_{ij}, \quad (\delta_x u)_{ij} = N \cdot \begin{cases} u_{ij} - u_{i-1,j}, & u_{ij} > 0, \\ u_{i+1,j} - u_{ij}, & u_{ij} \leq 0, \end{cases} \quad (10)$$

δ_y analogous and switching on the same centre value, ghost zeros on all four walls, and backward Euler in time, $r_n(u) = u - u^n + \Delta t(N(u) - \nu \Delta_h u)$. Projecting and row scaling by $\Lambda_\star = I + \Delta t \nu \Lambda$ gives, with $c = h_\theta(z) + C_q y$,

$$r_{w,n} = (I + \Delta t \nu \Lambda)^{-1} [B_0 c - B_0 c_n + \Delta t (\hat{N}(c) + \nu \Lambda B_0 c)], \quad (11)$$

where the diffusion term is exact and the row scaling is the diagonal of the linearised implicit operator in the modal basis, the modal form of the Helmholtz preconditioner the full-order Newton solver uses. Only $\hat{N}(c) \approx PN(Gc)$ requires a choice: dense evaluation at every interior node, the sampled rule $\hat{N}(c) = \sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(Gc)$ with the sign switch retained at the selected nodes. Time stepping uses a fixed substep count per output interval; each step is warm-started at the previous code, and the linear extrapolation is used instead when, and only when, its residual norm is smaller; the 4096² rows also try a quadratic predictor under the same rule. The runs above 1024² assemble the Jacobian analytically instead of by forward-mode products and clip the step; in the settings listed in Table B.4 they solve the normal system by Cholesky factorisation. In every Burgers arm above 1024² except the 2048² development accurate arm, the damping λ is carried from one step to the next rather than reset to λ_0 .

Block-damped step. With $J = [J_z \ J_y]$ the Jacobian of (11) in (z, y) , obtained in one forward-mode pass, each iteration solves

$$\begin{pmatrix} J^\top J + \lambda D_z + \varepsilon D_y \end{pmatrix} \begin{bmatrix} \delta z \\ \delta y \end{bmatrix} = -J^\top r_{w,n}, \quad \|\delta z\| \leq \Delta, \quad (12)$$

where $D_z = \text{diag}(\text{diag}(J_z^\top J_z), 0)$ carries the Levenberg–Marquardt damping on the latent block only, $D_y = \text{diag}(0, I)$ with a fixed small ridge ε , and Δ is the $q = 0$ trust radius. The correction step is therefore an undamped Gauss–Newton step on the current linearisation, not throttled by a radius calibrated for the latent code. The step is accepted only if the residual decreases; otherwise λ grows as in §A.6.

A.4 SECOND-ORDER INSTANCE: REFLECTIVE WAVES

For $u_{tt} = c^2 \Delta u$ the bank is mass-orthonormal and the R weak tests are the bank itself, so projection is Galerkin with stiffness $K = G^\top M_h A G$, M_h the diagonal mass matrix. Writing the coefficients as $a(z, y) = h_\theta(z) + C_q y$ and differentiating twice in time, $\ddot{a} = D h_\theta \ddot{z} + h''_\theta(z)[\dot{z}, \dot{z}] + C_q \ddot{y}$, the projected equation $\ddot{a} = -c^2 K a$ becomes a linear least-squares problem for the joint acceleration,

$$\begin{bmatrix} D h_\theta(z) & C_q \end{bmatrix} \begin{pmatrix} \ddot{z} \\ \ddot{y} \end{pmatrix} = -\left(c^2 K a + h''_\theta(z)[\dot{z}, \dot{z}]\right), \quad (13)$$

whose curvature term $h''_\theta[\dot{z}, \dot{z}]$ is a forward-over-forward derivative of the head. The first-order system in $(z, y, \dot{z}, \dot{y})$ is advanced with classical RK4 at a fixed step, one solve of (13) per stage; there is no implicit residual minimisation and no stationarity exit. Initial coordinates and velocities are obtained from the supplied displacement and velocity. The iterative full-order comparator uses implicit midpoint with CG. The error reported in Table 3 is the energy-state error over displacement and velocity.

A.5 EMPIRICAL QUADRATURE: THE FIT AND THE COUNTS

A fitted rule with support $\mathcal{S}$ and weights w is scored by the held-out relative error of the projected advection term,

$$\rho(u) = \frac{\|\sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(u) - PN(u)\|_2}{\|PN(u)\|_2}, \quad \rho_{\max} \leq 0.116 \text{ (primary bar)}, \quad \rho_{\max} \leq 0.06 \text{ (tight)}, \quad (14)$$

over the held-out reachable states described below. The primary bar is the held-out ρ of the original model’s $q = 0$ rule at the state that carries its largest first-interval error, measured in an earlier diagnostic study and fixed before any certification job ran; the tight bar was fixed before the EQ-ladder job of Table C.2. A candidate node set $\mathcal{C}$ is drawn once with a fixed seed. For each of n_{fit}

stored codes $c^{(\ell)}$ we evaluate the decoded state, form the per-node contributions $N_i(Gc^{(\ell)})$ and the exact full-grid target $PN(Gc^{(\ell)})$, and solve

$$\min_{w \geq 0} \|\mathcal{G} w - \beta\|_2, \quad \mathcal{G}_{(\ell,i),c} = P_{i,c} N_c(Gc^{(\ell)}), \quad \beta_{(\ell,i)} = [PN(Gc^{(\ell)})]_i, \quad (15)$$

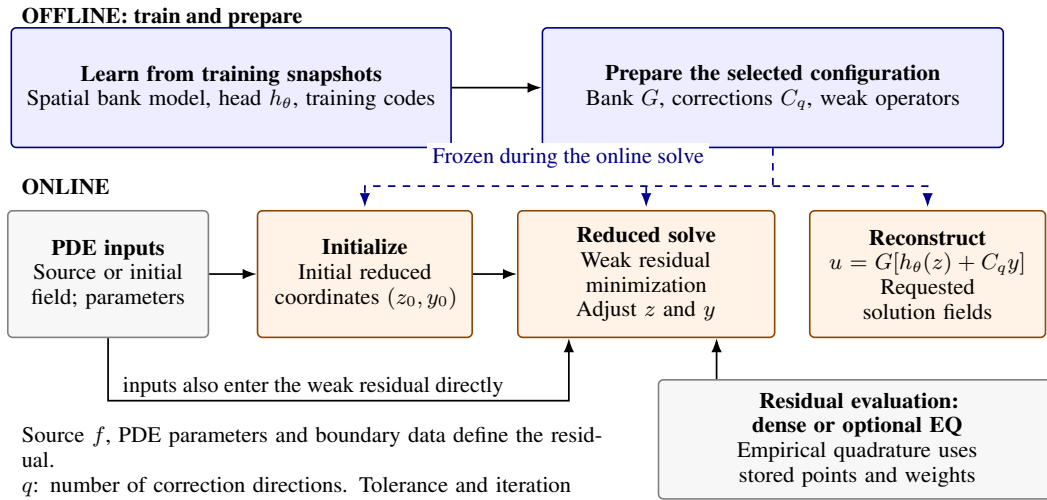
rows normalised by their Euclidean norms, support grown greedily by the Lawson–Hanson criterion with an exact non-negative refit after each addition and a hard cap of m nodes, padded to the requested count if the support saturates. Three counts are separated: the requested node count m , the achieved support, and the number of fit states n_{fit} . In the certification study the fit states are reachable states (states of the dense solver’s own converged per-step trajectory on fit trajectories), the held-out states are 512 reachable states per rung from certification trajectories disjoint from both the fit and the evaluation cases, and ρ of (14) is evaluated on those. A deployed rule is a stored triple: node indices, weights, and the cached $m \times 5 \times R$ bank block.

A.6 SOLVER CONSTANTS AND EXIT CODES

Damping starts at $\lambda_0 = 10^{-6}$ (Poisson, Burgers) or 10^{-4} (heat); $\lambda \leftarrow \max(\lambda/3, 10^{-12})$ on acceptance and $\min(10\lambda, \lambda_{\text{max}})$ on rejection; the trust radius is taken from the spread of the training codes. Exit codes differ per problem and are reported per problem. Burgers: 4 stationarity, 1 residual, 2 tiny step, 3 rejected at $\lambda \geq 10^{14}$, 0 budget. Poisson: 6 stationarity, 2 relative residual, 1 stall, 3 damping limit, 5 non-finite initial value, 0 budget. Heat: 1 stationarity, 2 tiny step, 3 damping limit, 4 non-finite, 0 budget. The heat criterion divides by $\|J\|_F$ only, applied to a residual already normalised by the target norm; Poisson and Burgers use (4) directly. An initial fit that meets its stopping rule is accepted as the starting state of the time loop; this acceptance rule was fixed before the evaluation jobs ran, and complete exit flags remain in the source records.

A.7 ARCHITECTURE AND ONLINE DATA FLOW

Figure A.1 shows the offline preparation and the online solve of every NM-ROM query in the paper.



G : frozen spatial bank; h_θ : frozen neural head; C_q : fixed correction directions; z : nonlinear coordinates; y : correction coefficients. Solid arrows show online flow, the direct input dependency of the residual, or an evaluation choice; dashed arrows show reuse of prepared components.

Figure A.1: Offline preparation (blue) and online solution (orange). The PDE input supplies both initialization and the weak residual. Linear-PDE corrections are eliminated analytically; time-dependent problems repeat the reduced step. EQ is used only for 2D Burgers.

B EXPERIMENTAL CONFIGURATION AND REPRODUCIBILITY

Tables B.2 and B.3 identify the two-dimensional studies and Table B.6 the three-dimensional ones. Within each study, training precedes evaluation and the selected bank, head and correction directions remain frozen. Deployment changes reduced coordinates, not network weights. Recorded training configurations and available offline costs are retained with the source evidence; unrecorded training costs are not inferred. The wide heat bank of Table 1 is a random-Fourier-feature coordinate network (32 features, scale 1.5, width 256, $R = 128$) trained for 80000 steps (learning rate 0.001, 64 states $\times$ 2048 points per batch, gradient clip 1, exact coefficient refit every 10000 steps), then a head of width 256 with $k = 8$ for 80000 steps (learning rate 0.0005, 128 states per batch), on 512 training draws at 64^2 with 16 validation draws; its validation projection floor is 0.11 %.

Table B.1: Training configurations of the frozen checkpoints, transcribed from the committed configuration and code files of each study (paths and SHA256 in evidence/training-configs-2026-09-21).

Model	Stage	Network	Optimiser, learning rate	Steps	Batch	Training data
Burgers 2D ($k = 16$, $R = 512$)	bank	128 Fourier features (scale 4), width 1024, 2 layers	AdamW, wd 10^{-5} , warm-up + cosine $10^{-3} \rightarrow 10^{-5}$	300000	all states $\times$ 4096 points	16384 states of 576 trajectories
	head	width 512, 2 layers	Adam, warm-up + cosine $10^{-3} \rightarrow 10^{-5}$	200000	4096 states	131072 states of 4608 trajectories
Poisson 2D ($k = 32$, $R = 512$)	bank	64 Fourier features (scale 4), width R , 2 layers	Adam, four phases at 10^{-3} , 3×10^{-4} , 3×10^{-4} , 10^{-4}	100000, 40000, 30000, 30000	64 sources $\times$ 4096 points	2611 fit / 461 validation sources
Poisson 2D ($k = 16$, $R = 128$)	head	width 128, 2 layers	Adam, 10^{-3}	150000 (full batch)	all	as bank
	bank + head	64 Fourier features, g and h width 128, 2 layers	Adam; staged from an $R = 64$ model (bank, head, joint at 3×10^{-4} , 3×10^{-4} , 10^{-4}) as Poisson $k = 32$	40000, 30000, 20000 after widening	64 sources $\times$ 4096 points	512 training sources
L-shape ($k = 16$, $R = 512$)	bank / head	64 Fourier features (scale 4), 2 layers; head width 128, 2 layers	Adam, cosine (floor 0.03); 10^{-3}	100000, 40000, 30000, 30000; head 150000	64 sources $\times$ 4096 points	2611 fit / 461 validation sources
Poisson 3D ($k = 16$, $R = 128$)	bank / head	64 Fourier features (scale 1.5), widths 256; head width 256 + skip	Adam, cosine (floor 0.03); 10^{-3} bank, 5×10^{-4} head	150000 / 100000	64 states $\times$ 2048 points / 128 states	512 training / 16 validation fields

Full-order and solver details. The Burgers FOM is Newton’s method with BiCGStab inner solves (at most 200 iterations, Jacobian–vector products by forward-mode differentiation) preconditioned by the exact inverse of the implicit Helmholtz operator $(I + \Delta t \nu \lambda)^{-1}$ (advection dropped); Newton stops when the residual norm falls below the nonlinear tolerance times the previous state’s norm, or after 20 iterations, and always takes the full step (no line search or damping). Each named setting fixes Δt , the nonlinear and the linear tolerance. On the L-shape the vanishing factor is $4 \min(x, y, d_{\text{right}}, d_{\text{top}}, d_{\text{corner}})$ with the distances to the outer right and top edges and to the re-entrant corner, and the test space is the M lowest eigenvectors of the five-point operator on the masked grid, computed by shift-invert Lanczos and scaled as $P = \Lambda^{-1} \Phi^T$ ($M = 257$; $M = 1024$ for the $q = R$ solve). The quadratic predictor of the 4096^2 Burgers rows forms the linear extrapolation $2w_n - w_{n-1}$ and, from the third step on, the quadratic one $3w_n - 3w_{n-1} + w_{n-2}$, evaluates the residual norm of the current code and both candidates in one batched call and starts the solve from the smallest.

Three-dimensional heat, new bank. The bank of the Heat (new bank) rows ($R = 320$) was trained with a code-free variable-projection trainer that eliminates the coefficients exactly; its projection floor is 0.06 %. Its arms were fixed by a rule registered before the sealed cohort was opened.

B.1 HEADLINE SETTINGS, TIMES AND TIMING PROTOCOL

Table B.4 gives the settings and measured times behind every row of Table 1. Every cost and error come from the same invocation. Timings use GPU warm-up and synchronisation, double precision and the highest matrix-multiplication precision; all repetitions, including outliers, enter the median, and no time is borrowed from another allocation. Requested full fields and initialisation are charged; host transfer is included only where the table says complete query. For the held-out Burgers rows, an independent restricted-grid recomputation of the errors disagreed with the full-grid value by more than 5 % on $5/384$ (2048^2) and $18/1152$ (4096^2) case-arm rows; per-arm cohort worsts agree within 1.2 % and the full-grid recomputation is exact. For Burgers at 256^2 no quadrature rule passed the pre-registered re-draw procedure (its pick failed the confirmation draw), so Table 1 has no confirmed-rule row there; a post-hoc follow-up rule with an exact first time step passes all 12 draws of two jobs and gives 0.520 % at $0.085 \times$ against the same Newton–BiCGStab rule. For Burgers at 4096^2 , copying the six double-precision output fields to the host adds about 246 ms to every arm. The Poisson and three-dimensional heat CG comparators are unpreconditioned; heat CG is warm-started from

Table B.2: Problem specification. Cohort and reduced sizes are read from the run configurations where recorded. The Burgers sealed cohort has been opened and is reported in Table C.1; other rows describe their recorded development and validation cohorts.

PDE	equation, domain, boundary	meshes	time stepping	reduced sizes	reference	cohorts
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 – 4096^2	$\Delta t = 0.005$, backward Euler, sign- upwind	$k = 16$, $R = 512$	refined 4096^2 , $\Delta t = 0.00015625$	6 development cases; 32 validation; 64 held-out at 2048^2 , 4096^2 ; sealed co- hort opened once
Poisson 2D	$-\Delta u = f$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 – 4096^2	none (elliptic)	$k = 16$, $R = 128$ (original); $k = 32$, $R = 512$	exact discrete solution; 2048^2 refinement	12 development sources
Heat 2D	$u_t = \kappa \Delta u$, $(0, 1)^2$	64^2 – 1024^2	Crank–Nicolson	$k = 8$, $R = 32$	refined-grid reference ($1024^2/2048^2$ pair); error includes discreti- sation	12 development cases, 3 repetitions; measured in job 3529772; checkpoint lineage: earlier cell, job 3511417
Heat 2D, wide bank	$u_t = \kappa \Delta u$, $(0, 1)^2$	1024^2 – 4096^2	Crank–Nicolson; batched exact- propagator fit	$k = 8$, $R = 128$	exact same-grid propa- gation	12 development cases; 16 sealed held-out cases, opened once
Wave 2D (reflec- tive)	$u_{tt} = c^2 \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	64^2 , 256^2 , 1024^2	RK4 on the manifold; exact modal propaga- tion for the bank	$k = 32$, $R = 64$	exact modal solution	8 development cases
Poisson, L-shape	$-\Delta u = f$, $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	256^2 – 2048^2	none	$k \in \{16, 32\}$, $R \in \{256, 512, 514\}$	converged discrete so- lution	3072 / 256 / 32 sources (train / selection / develop- ment)

Table B.3: Sampled problem families, transcribed from the generator sources named in the last column (code constants, not run outputs).

PDE	equation and boundary	sampled family (transcribed from the generator source)	source
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$u_0 = a \exp(- x - c ^2/2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim U(0.05, 0.20)$, $a \sim U(0.5, 2)$; $\nu \sim$ $\log U(0.01, 0.1)$; outputs $t \in \{0.05, \dots, 0.25\}$	experiments/ mr-burgers2d/ engines.py: params_draw
Poisson 2D	$-\Delta u = f$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$f = a \exp(- x - c ^2/2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim$ $\log U(0.02, 0.1)$, $a \sim U(0.5, 2)$	multistage- precision/ms_ parametric.py: sample_params
Poisson, L- shape	same source family on $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	as above; sources centred in the removed quadrant rejected	experiments/ lshape
Heat 2D	$u_t = \kappa \Delta u$ on $(0, 1)^2$, $\kappa = 0.02$ fixed	polynomial-boundary Gaussian family (sin- gle_bc_poly_gaussian_v1); Crank–Nicolson $\Delta t = 0.025$	checkpoint expanded_ seed790715 (lineage: linear-bank cell, job 3511417); printed CG measurements: job 3529772
Wave 2D (re- flective)	$u_{tt} = c^2 \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	compact bump $\times$ Gaussian: half-widths $s_i \sim U(0.36, 0.42)$, centre $c_i \sim$ $U(s_i + 0.025, 1 - s_i - 0.025)$, amplitude $\sim U(0.7, 1.3)$, $\sigma_i \sim U(0.12, 0.16)$, advective velocity $v_i \sim$ $U(-0.5, 0.5)$ (zero every fourth case); speed $c \sim U(0.85, 1.15)$	experiments/ multiresolution- wave/audit_ dynamics.py: parameter_rows

the previous time level. The Heat2D rows use checkpoint `expanded_seed790715`; Table B.2 separates that checkpoint’s lineage from the job that produced the printed measurements.

Model and table annotations. Heat (wide bank) is a separately trained frozen model with a 128-function bank ($q = 32$ accurate); over evolved times its accurate error is 0.22 % (0.21 % batched; the batched fit: Table B.4); Heat (new bank) ($q = 288$ accurate; Table B.6): at most 0.086 % (0.030 % batched) over evolved times; ⁿ 1 solve (one step of one case) missed its stationarity rule, so no speedup is given.

Burgers rule identities. Burgers accurate quadrature: ^s stored rule of job 3780164 ($m = 2560$; not the $q = 256$ rule of Table C.2), not confirmed on re-draws: it fails the confirmation draw at 256^2 ; ^x its nodes with weights refit at 512^2 ($m = 2438$), one held-out draw, not re-drawn (with unrefit weights they pass 4 of 5 re-draws); ^l 63×63 lattice rule: at 2048^2 it clears the re-draw bar by only 4×10^{-4} ($\rho_{\max} = 0.1156$ vs. 0.116). At 4096^2 , a later confirmation draw fails the bar. A variant with an exact first step passes the check but loses the speed advantage; ^v confirmed by the pre-registered re-draw procedure (1024^2 : thin margin, $\rho_{\max} = 0.1157$; none passes at 256^2 , Appendix B.1); ^d dense residual. The earlier Burgers model has one setting and may exit on a stall; its rule-selected FOM is the relaxed Newton setting ($10^{-2}/0.5$), the FOM setting can differ between meshes of a series. ^h 64 held-out cases never used for selection (not the sealed final cohort); at 2048^2 the development-chosen $M = 544$ setting was not run on them, so the $M = 1088$ setting is shown.

The separate bank follow-up also missed its combined held-out accuracy and speed target. The committed follow-up handoff is retained with its source hash in `evidence/readability-2026-09-22`.

Three-dimensional Poisson ($-\Delta u = f$, Gaussian sources), heat ($u_t = \kappa \Delta u$, Gaussian initial fields) and scalar Burgers ($u_t + u(u_x + u_y + u_z) = \nu \Delta u$, backward Euler, sign-upwind) are posed on the unit cube with zero Dirichlet walls and extend the operators of Appendix A across three axes. Navier–Stokes uses periodic vector fields, an incompressibility projection and the CNAB2 integrator; its FOM is not a CG solve. All four three-dimensional problems are evaluated on held-out final cohorts whose settings were frozen beforehand.

Source records. The repository retains configurations, checkpoints, case-level outputs, timing repetitions and audits. The manifests in `tables/` identify the source commits and hashes for each generated value; `headline-provenance.json` covers the main comparison tables. The Burgers training and evaluation parameters were regenerated and checked for overlap: none was found. The check and its output are in `evidence/burgers-train-eval-overlap-2026-09-21`.

B.2 REPRODUCED NM-ROM BASELINE

The shallow masked-autoencoder NM-LSPG of Kim et al. (2022) passed its reproduction gate on the third registered attempt on the published Burgers benchmark: median 1.45 % against a 1.5 % gate (sigmoid activation). Its hyper-reduction did not pass the reproduction gate. Table 2 therefore uses dense residuals for every method. Our reference path uses tolerance 10^{-6} and $M = 4(k + q)$. The published encoder width, $2n$, exceeds device memory. At 512^2 , the width-capped encoder (4096) trained for only 229–231 epochs within 4800 s. Its error thus reflects a training constraint as well as the representation and solve. Exploratory timings and failed hyper-reduction runs remain in the repository evidence.

C VALIDATION OF THE CORRECTION AND QUADRATURE STUDIES

Reading the validation tables. Correction rank q is the number of added coefficient directions; M is the number of weak test modes and m the quadrature node count. A scheduled ladder changes M with q ; the main fixed-test-space study holds M constant. A sealed cohort is opened only after choices are frozen. Confirmed quadrature passes every independent reconstruction of the rule; marginal means some reconstructions fail. These statuses concern rule construction, not just the error of one selected trajectory. Same-grid error and error against a refined reference measure different quantities.

Table B.4: Settings and median times for exactly the rows of Table 1, using the same FOM setting and allocation. Settings give q , or $q/M/m$ where recorded; D means dense and * an exact first step. Times include initialization, solution and requested device fields; ^c includes host transfers as well. The last column gives complete-query speedups (accurate / fast) where separately measured. ⁿ marks a nonstationary solve whose speedup is omitted from Table 1. The batched heat fit treats output times independently using exact weak-moment propagation.

Problem	Mesh	Acc. setting	Fast setting	Acc. ms	Fast ms	FOM ms	Complete-query S
Poisson 2D	256 ²	256	0	7.12	6.70	22.29	—
Poisson 2D	1024 ²	256	0	4.32	3.94	60.02	—
Poisson 2D	2048 ²	256	0	5.47	5.35	397.80	28.7 / 28.8
Poisson 2D	4096 ²	256	0	16.92	16.74	1954.80	32.7 / 33.2
Poisson, L-shape 2D ^c	256 ²	64	0	3.03	2.88	11.64	—
Poisson, L-shape 2D ^c	512 ²	64	0	4.80	4.72	29.13	—
Poisson, L-shape 2D ^c	1024 ²	128	0	5.81	5.67	48.70	—
Poisson, L-shape 2D ^c	2048 ²	128	0	18.75	18.66	318.22	—
Heat 2D	64 ²	—	single	—	11.23	2.50	—
Heat 2D	256 ²	—	single	—	11.34	6.94	—
Heat 2D	1024 ²	—	single	—	12.21	59.18	—
Heat (wide bank) 2D	1024 ²	32	0	27.33	27.77	43.51	—
Heat (wide bank) 2D	2048 ²	32	0	27.97	28.42	272.55	—
Heat (wide bank) 2D	4096 ²	32	0	31.68	32.19	1139.83	—
Heat (wide bank, batched fit) 2D	1024 ²	32	0	3.13	3.36	43.51	—
Heat (wide bank, batched fit) 2D	2048 ²	32	0	4.54	4.78	272.55	—
Heat (wide bank, batched fit) 2D	4096 ²	32	0	11.03	11.33	1139.83	—
Burgers 2D	256 ²	256/1088/2560	0/64/1024	746.02	40.36	31.79	—
Burgers 2D	512 ²	256/1088/2438	0/64/922	783.33	40.49	52.92	—
Burgers 2D	1024 ²	256/1088/D	0/64/934	22053.85	40.27	81.31	—
Burgers 2D	2048 ²	256/544/3969	0/64/1024	124.15	29.48	122.76	0.99 / 2.01
Burgers 2D	4096 ²	256/1088/3969	0/64/1024	107.51	40.73	523.85	2.18 / 2.68
Burgers (held-out cases) 2D	2048 ²	256/1088/3969	0/64/1024	114.91	29.19	164.43	1.28 / 2.49
Burgers (held-out cases) 2D	4096 ²	256/1088/3969	0/64/1024	99.96	40.60	537.39	2.28 / 2.74
Burgers, confirmed rule 2D	512 ²	256/1088/3969*	0/64/1024	194.97	23.62	17.74	—
Burgers, confirmed rule 2D	1024 ²	256/1088/3969	0/64/1024	108.58	25.00	34.69	—
Burgers (earlier model) 2D	1024 ²	—	single	—	41.68	68.04	—
Poisson 3D	32 ³	96	0	2.63	2.49	2.47	—
Poisson 3D	64 ³	96	0	3.36	3.22	4.46	—
Poisson (dev. sources) 3D	128 ³	96	0	1.86	1.80	12.55	2.70 / 2.66
Poisson (dev. sources) 3D	256 ³	96	0	6.08	5.95	140.87	4.18 / 4.21
Heat (new bank) 3D	32 ³	288	0	58.16	25.41	3.33	—
Heat (new bank) 3D	64 ³	288	0	58.69 ⁿ	25.45	5.23	—
Heat (new bank) 3D	128 ³	288	0	62.03	28.25	17.44	—
Heat (new bank, batched fit) 3D	32 ³	288	0	5.36	5.88	3.33	—
Heat (new bank, batched fit) 3D	64 ³	288	0	5.54	6.14	5.23	—
Heat (new bank, batched fit) 3D	128 ³	288	0	7.84	8.30	17.44	—

Table B.5: Burgers 2D correction settings from one frozen model at 256², test count fixed at $M = 1088$, dense residual, one allocation. Errors are worst evolved same-grid percentages on development cases; times are paired median GPU query times. All rows meet the stopping rule.

Correction rank q	Relative L^2 error (%)	GPU query time (ms)
0	1.2657	848.0
64	1.0593	1359.4
128	0.8711	1886.5
256	0.5194	4377.9

Table B.6: Three-dimensional configurations, read from the run records. Domains are the unit cube with homogeneous Dirichlet data, except Navier–Stokes, which is periodic. Burgers and Navier–Stokes appear in Table 3.

Problem	Mesh	Cases	Correction ranks	Named FOM	Residual
Burgers 3D	32 ³	32 final	0, 192	Newton–BiCGstab, $\Delta t = 0.01$	dense
Poisson 3D	32 ³ , 64 ³	64 final	0, 32, 96 ($k = 16$)	CG, rtol 10^{-2} , no preconditioner	dense
Heat 3D (new bank)	32 ³ , 64 ³ , 128 ³	64 sealed final (all times)	0, 288 ($k = 32$, $R = 320$)	CN–CG, setting per row by the rule; named $\Delta t = 0.025$, rtol 10^{-6}	exact (linear)
Navier–Stokes 3D	32 ³ periodic	32 final	0, 32, 64, 128, 256 ($k = 64$, $R = 1536$, $M = 2048$)	CNAB2, $\Delta t = 0.01$	dense

Table C.1: The sealed cohort (job 3804465, NVIDIA A100-PCIE-40GB), opened once after every choice was frozen. Top: per rung of the dense $M = 4(k + q)$ ladder, the original model’s sealed worst evolved error and device time, the seed mean $\pm$ sample standard deviation on the sealed and the development cohort, the pre-registered sealed-over-development ratio of seed means (raw and difficulty-normalised) and the converged count over the 4 checkpoints. Bottom: per-checkpoint verdicts. The original model’s $q = 0$ rung is a single sealed case converged to a wrong branch (gradient exit, zero budget exits); the pre-registered ratio criterion fails at $q = 0$ for the original model alone (5.35) and the convergence criterion fails on the second retrained seed at $q = 64$ (3 budget exits). Top-rank sealed errors per checkpoint: original 0.68 %, first retrained seed 0.61 %, second 0.59 %, third 0.66 %.

q	M	original model sealed %	ms	seeds sealed %	seeds development %	sealed / dev	normalised	converged
0	64	10.1120	375.1	2.8023 ± 0.6357	2.0002 ± 0.4986	1.40	1.41	4 of 4
16	128	1.4617	480.8	1.7305 ± 0.1297	1.5988 ± 0.0995	1.08	1.09	4 of 4
32	192	1.4458	540.5	1.5044 ± 0.1422	1.3727 ± 0.1314	1.10	1.14	4 of 4
64	320	1.3252	754.5	1.3528 ± 0.1099	1.2130 ± 0.0580	1.12	1.13	3 of 4
128	576	1.0964	1460.4	1.0461 ± 0.0619	0.9562 ± 0.0264	1.09	1.17	4 of 4
256	1088	0.6789	6725.5	0.6188 ± 0.0345	0.4987 ± 0.1164	1.24	1.20	4 of 4

checkpoint	job	monotone (evolved)	monotone (all-times)	every setting converged	error span	cost span	secondary criterion
original	3804465	yes	yes	yes	$14.89 \times$	$17.93 \times$	yes
first seed	3804465	yes	yes	yes	$3.52 \times$	$13.25 \times$	yes
second seed	3804465	yes	yes	no	$4.81 \times$	$13.17 \times$	no
third seed	3804465	yes	yes	yes	$5.22 \times$	$14.62 \times$	yes

Table C.2: The EQ ladder with the cheapest rule passing the primary bar in its draw per rung, timed in one allocation (job 3780164, A100 80 GB), with its same-job dense twins, and the construction status from the four-draw replication (job 3783811): confirmed = every re-draw passes, marginal = only the listed fraction does. Errors are as measured with these rules (SHA256-identified); they are not properties of the construction above $q = 32$. Ladder rows are read from the study’s final summary record.

q	m	fit states	$\rho_{\max}$ (this draw)	construction status	EQ evolved %	EQ all %	EQ ms	dense evolved %	dense ms	EQ/dense cost
0	1024	64	0.0153	confirmed (3 of 3 re-draws)	1.8891	2.5629	59.1	1.8890	292.9	0.202
16	1024	64	0.0935	confirmed (3 of 3 re-draws)	1.4270	2.4806	80.6	—	—	—
32	1024	42	0.0533	confirmed (2 of 2 re-draws)	1.2493	2.3534	97.7	—	—	—
64	1024	25	0.0531	marginal (2 of 6 draws pass)	1.2275	2.1489	120.6	1.0843	624.9	0.193
128	2048	64	0.0669	marginal (4 of 5 draws pass)	0.8936	1.8116	246.9	0.8930	1190.5	0.207
256	2048	64	0.1074	marginal (1 of 5 draws pass)	0.5389	0.9053	722.2	0.5194	3915.1	0.184

Table C.3: Solver-side controls on Burgers 2D, 32 validation cases, one allocation: stopping tolerance, EQ node count m and iteration cap, with full-order settings for scale. Error: worst over cases and output times of the L^2 error against the refined reference, relative to the initial-field norm, so the full-order rows show their discretisation error.

setting	worst % (32 validation)	device ms	early-stopped
EQ $m=256$, tol 10^{-6} , cap 180	7.245	51.0	0/96
EQ $m=512$, tol 10^{-8}	6.712	64.7	0/96
EQ $m=512$, tol 10^{-3}	6.701	41.6	0/96
EQ $m=512$, cap 2 (early-stopped)	90.324	28.4	96/96
FOM Newton 10^{-2} , $\Delta t=0.01$	6.807	9.9	0/96
FOM Newton 10^{-2} , $\Delta t=0.005$	35.357	17.4	0/96
FOM Newton 10^{-4} , $\Delta t=0.005$	6.171	22.4	0/96
FOM Newton 10^{-6} , $\Delta t=0.005$	6.172	88.3	0/96